Python Engineering - Reference Guide

Class 12 - Engineering Toolkit: NumPy functions in EX20

Reference guide for the important functions used in EX20: what they do, what they receive, what they return, and a small example.

np.linalg.norm()

Aspect	Description
What it does	Computes vector/matrix norm or magnitude.
What it receives	numerical array
What it returns	scalar by default

```
np.linalg.norm([3,4]) -> 5.0
```

np.isclose()

Aspect	Description
What it does	Compares two scalars using tolerances.
What it receives	two numbers + optional tolerances
What it returns	True/False

```
np.isclose(norm, 0.0)
```

np.dot()

Aspect	Description
What it does	Computes dot product for 1D vectors and related products for other dimensions.
What it receives	two compatible arrays
What it returns	scalar or array

```
np.dot([3,4],[1,0]) -> 3
```

np.clip()

Aspect	Description
What it does	Limits values to a specified interval.
What it receives	array/scalar, minimum, maximum
What it returns	clipped value/array

```
np.clip(1.0000001,-1,1) -> 1
```

np.arccos()

Aspect	Description
What it does	Computes inverse cosine element-wise.
What it receives	value in [-1,1]
What it returns	angle in radians

```
np.arccos(0.0) -> pi/2
```

np.rad2deg()

Aspect	Description
What it does	Converts radians to degrees.
What it receives	angle/array in radians
What it returns	degrees

```
np.rad2deg(pi/2) -> 90
```

np.eye()

Aspect	Description
What it does	Creates an identity matrix.
What it receives	dimension N
What it returns	NxN matrix

```
np.eye(2) -> [[1,0],[0,1]]
```

np.linalg.det()

Aspect	Description
What it does	Computes determinant of a square matrix.
What it receives	NxN matrix
What it returns	scalar

```
det(R) approx 1
```

np.allclose()

Aspect	Description
What it does	Checks approximate equality of all elements.
What it receives	two arrays/values
What it returns	True/False

```
np.allclose(R.T@R, I)
```

Numerical-safety summary

In numerical computing, exact float equality may be unreliable. That is why we use np.isclose() and np.allclose(). np.clip() also protects np.arccos() from tiny round-off excursions such as 1.0000000001 or -1.0000000001.